Increasing Cross-Layer Scope & Multi-Tool Verification

LAYER 3 — AI-Native System Design — Cross-Boundary Closed-Loop Orchestration (Architecture 2.0 Moonshot)

Operational Scope
- Cross-boundary closed-loop multi-tool search
- Workload ↔ Compiler ↔ RTL ↔ Physical ↔ Formal
- Co-explores microarchitecture & PDK signoff

Key Technical Mechanisms
- Multi-level IRs (CIRCT / MLIR dialect lowering)
- Rejectable multi-fidelity check harnesses
- Cryptographic lineage & design-loop cards

Governance & Signoff Boundary
- In-Order Human Commitment Authority
- Multi-property formal & physical signoff gates
- Governs daily code churn & tool dispersion

LAYER 2 — AI-Driven Subsystem Optimization — Autonomous Deep Local Search Inside a Fixed Container

Operational Scope
- Autonomous search in a single subsystem
- Macro floorplanning & kernel autotuning
- DRAM controller timing & DSE sweeps

Key Technical Mechanisms
- Reinforcement learning (e.g. AlphaChip)
- Bayesian optimization & conformal models
- Domain-specific surrogate cost models

Governance & Signoff Boundary
- Bounding container frozen by architect
- Cannot alter cross-stack interfaces
- Bounded by Waterbed Effect & Amdahl limits

LAYER 1 — AI-Assisted Engineering — Human-in-the-Loop Point Task Speedup Within Existing Workflows

Operational Scope
- Accelerates isolated drafting & typing
- RTL module generation & testbench templates
- Timing log parsing & script generation

Key Technical Mechanisms
- Open-loop foundation models & copilots
- Unstructured natural language & tokens
- Syntax-level completion & code search

Governance & Signoff Boundary
- Human acts as manual integration bus
- Generates syntax without signoff proof
- Severe physical attrition (<1% tapeout-ready)

The Architecture 2.0 Premise: Point assistance (Layer 1) and subsystem tuning (Layer 2) accelerate components, but only AI-native system orchestration (Layer 3) carries compact intent through real tools to physically realizable silicon.